

STANDARDIZED OPTIMAL DESIGNS FOR BINARY RESPONSE EXPERIMENTS

Holger Dette and Michael Sahn

Fakultät und Institut für Mathematik, Ruhr-Universität Bochum,
Universitätsstr. 150, Gebäude NA, 44780 Bochum, Germany

Key words: Binary response model; logistic regression; probit model; standardized optimal design; A- and E-optimal design.

Summary: In this note we consider the optimal design problem for Binary response models with two parameters. It is demonstrated that some of the commonly used optimality criteria (A-, E- ...) yield rather awkward designs of the experiment, in the sense that the optimal designs degenerate to one-point designs which are useless for statistical inference in models with two parameters.

Based on these observations we propose a standardization of the optimality criterion in order to eliminate different scales of the elements in the Fisher information matrix caused by different sizes of the parameters in the model. The proposed criteria are not affected by arbitrary scale transformations. It is shown that the determination of a standardized optimal design can be reduced to the corresponding non-standardized problem. For the logistic and probit regression model standardized and non-standardized A- and E-optimal designs are calculated explicitly and their different properties are illustrated.

1. Introduction

We consider the common Binary response model where a subject is administered a stimulus at a dose level x . The response X is a binary random variable with probability of success $p(x, \theta)$ where θ is an unknown

multidimensional parameter, $X \sim \text{Bin}(1, p(x, \theta))$. Assume that the number of responses at dose levels $x_1, \dots, x_n$ are independent random variables $B_1, \dots, B_n$ such that $B_i \sim \text{Bin}(1, p(x_i, \theta))$, ($i = 1, \dots, n$), then the log-likelihood is given by

$$\sum_{i=1}^n [B_i \log p(x_i, \theta) + (1 - B_i) \log(1 - p(x_i, \theta))].$$

In this paper we restrict ourselves to two different parameterizations of a two-parameter Binary response model,

$$p_1(x, \theta) = F(a + bx) \quad \theta = (a, b), \quad x \in \mathbb{R}, \quad (1.1)$$

$$p_2(x, \theta) = F(\beta(x - \mu)) \quad \theta = (\mu, \beta), \quad x \in \mathbb{R}, \quad (1.2)$$

where F is a known distribution function with symmetric density function f .

In these models the Fisher information for the parameter θ at a point x is given by

$$I_1(x, \theta) = h(a + bx) \begin{pmatrix} 1 & x \\ x & x^2 \end{pmatrix},$$

$$I_2(x, \theta) = h(\beta(x - \mu)) \begin{pmatrix} \beta^2 & -\beta(x - \mu) \\ -\beta(x - \mu) & (x - \mu)^2 \end{pmatrix}$$

respectively, where the function h is defined as

$$h(x) = \frac{f^2(x)}{F(x)(1 - F(x))}. \quad (1.3)$$

A design is a probability measure ξ with finite support which requires the observations to be taken at the support points of the measure in proportion to the masses of the measure corresponding to the points. The Fisher information matrix of the design is defined by

$$I_1(\xi, \theta) = \int h(a + bx) \begin{pmatrix} 1 & x \\ x & x^2 \end{pmatrix} d\xi(x),$$

$$I_2(\xi, \theta) = \int h(\beta(x - \mu)) \begin{pmatrix} \beta^2 & -\beta(x - \mu) \\ -\beta(x - \mu) & (x - \mu)^2 \end{pmatrix} d\xi(x),$$

and an optimal design maximizes a real-valued function, say Φ , of the Fisher information matrix which is called optimality criterion. It is well known that in a Binary response model the optimal design depends on the unknown parameter θ . Following Chernoff (1953) numerous authors assume that an initial guess of θ is available and determine so-called "locally" optimal designs. An alternative is to assume a prior distribution for θ which leads to Bayesian optimality criteria (see Chaloner and Larntz, 1989, Mukhopadhyay and Haines, 1995, Haines, 1995) or to use sequential designs (see e.g. Ford and Silvey, 1980 or Wu, 1985).

The local optimal design problem for estimating a percentile in a binary response model has been discussed by Wu (1988), Kalish (1990). If the estimation of both parameters in (1.1) or (1.2) is of equal interest various optimality criteria have been investigated in the literature (see Haines, 1992, Abdelbasit & Plackett, 1983, Minkin, 1987, Sitter & Wu, 1993). Typical examples are D-, A- and E-optimality which entail minimizing the determinant, trace and maximum eigenvalue of the inverse Fisher information matrix and are special cases of Kiefer's class of Φ_p -criteria (see Kiefer, 1974). A D-optimal design minimizes the (first-order approximation of the) volume of the ellipsoid of concentration for the parameter θ . Similarly, the A-optimal design minimizes the sum of the variances of the parameter estimates, while

E-optimality is concerned with minimizing the maximum variance over all estimated, normalized linear combinations of the parameter.

The present paper has two purposes. On the one hand we present and compare A- and E-optimal designs for the different parameterizations (1.1) and (1.2) in some Binary response models. On the other hand we illustrate some practical drawbacks of the commonly used optimality criteria in these models. More precisely, it is shown that in some cases Kiefer's Φ_p -criteria will lead to locally optimal designs nearly concentrated on one point. Because such designs are not desirable from a practical point of view we propose the application of standardized optimal designs (see Dette, 1997) in the setup of Binary response models. In this case the design problems can be reduced to a *canonical form*, similar as in Ford, Torsney and Wu (1992), such that the optimization is independent of the unknown parameters. The approach is illustrated by considering the A- and E-criterion for the logistic and the probit regression model. Section 2 discusses some practical drawbacks of the optimal designs in the logistic regression, when the parameter of the slope is relatively small. It is demonstrated that in this case Φ_p -optimal designs ($p \neq 0$) degenerate to one-point designs which cannot be used for the estimation of both parameters. Section 3 introduces the concept of standardized optimality in the context of Binary response models with two parameters. Some concrete examples for the logistic and the probit regression are given in section 4. Finally some conclusions, extensions and technical proofs are given in section 5 and in the appendix.

2. Optimal designs for logistic regression

The logistic regression model is obtained by using the function

$$F(t) = (1 + e^{-t})^{-1} \quad (2.1)$$

in (1.1) and (1.2). For the parameterization (1.1) E-optimal designs were calculated by Dette and Haines (1994) using a geometric approach and are listed in the front line of table 2.1. The E-optimal designs for the parameterization (1.2) can be calculated in a completely analogous manner and are listed in the second line of table 2.1. Note that for both parameterizations the optimal design allocates the observations at two points which are symmetric with respect to the LD50. However, there are remarkable differences in the weights of the optimal designs. While in parameterization (1.2) the experimenter is advised to take half the observations in each of the two points, in parameterization (1.1) the weights of the optimal design are not equal (except for the case $a = 0$ or for large values of the slope parameter b). We now consider the situation where the slope in the logistic regression is relatively small. Intuitively, a good design for a model with a small slope should have its support points spread out as far as possible. However, we observe from table 2.1 that the E-optimal designs in a logistic regression with a small slope are almost concentrated in one point. For example, the E-optimal design for the parameterization (1.2) advises the experimenter to take all observations in a neighbourhood of the LD50 μ and does not allow any statistical inference regarding the slope β . For parameterization (1.1) the situation is similar. For example, if $|a| < c$, b sufficiently small, the design reduces to a Dirac measure at the origin. As an example that these phenomena appear even for a moderate size of the slope,

consider the parameterization (1.1) for $b = 0.2$. If $a = 1$, then the E-optimal design concentrates 99.03% of the observations at the point 0.099. On the other hand, if $a = 2.5$ the E-optimal design advises the experimenter to take 97.67% of the observations at the point -0.505. From a practical point of view these designs are of course not very useful.

Note that a small value of the slope can always be caused by a change of the units of measurement. For example, if the explanatory variable is measured in litre (l) and the "true" parameters for the logistic regression (1.2) are $\beta = 1$ and $\mu = 0$, the E-optimal design has equal masses at the points -1(l) and 1(l). However, a change in the units of measurement from (l) to millilitre (ml) produces an E-optimal design with equal masses at the points -0.001 (ml) and 0.001 (ml).

In order to give a heuristic explanation of these phenomena we remark that the optimal designs can be calculated by maximizing the corresponding optimality criteria for the matrices

$$I_1(\eta, \theta) = \int h(w) \begin{pmatrix} 1 & \frac{w-a}{b} \\ \frac{w-a}{b} & \frac{(w-a)^2}{b^2} \end{pmatrix} d\eta(w) \quad (w = a + bx) \quad (2.2)$$

$$I_2(\eta, \theta) = \int h(z) \begin{pmatrix} \beta^2 & -z \\ -z & z^2/\beta^2 \end{pmatrix} d\eta(z) \quad (z = \beta(x - \mu)) \quad (2.3)$$

in a first step and transforming the optimal designs via $X = (w - a)/b$ and $x = \mu + z/\beta$ in a second step. A calculation of the inverses of matrices (2.2) and (2.3) shows that for a small slope the element in the position (1, 1) will dominate the other elements in the inverse of the Fisher information matrix. Consequently we expect the E-optimal designs very close to the c-optimal designs for $c = (1, 0)^T$. However, in many cases these c-optimal

designs are one-point designs (see Wu, 1988, Ford, Torsney & Wu, 1992). For example, it follows from Elfving's theorem (Elfving, 1952) that in the Binary response model (1.2) the optimal design for estimating μ is a one-point design whenever $\sup_t h(t) = h(0) < \infty$, which is satisfied for many of the commonly used models. A similar observation can be made for large values of the slope parameter. In this case the variance of the estimator for the slope will dominate the other elements. Consequently the E-optimal design will not differ very much from the optimal design for estimating the slope, which, however, is not a too serious problem since the optimal design for estimating the slope is usually a fairly good design for estimating all parameters in the Binary response model.

Table 2.1. E-optimal designs for the logistic regression model (2.1) using two different parameterizations (1.1) and (1.2). Here $r = \sqrt{a^2 + b^2}$ and c is the unique positive solution of the equation $2 + u + 2e^u - ue^u = 0$.

$$[c \approx 2.39936]$$

	$r^2 = a^2 + b^2 \geq c^2$	$r^2 = a^2 + b^2 < c^2$
(1.1)	$\begin{pmatrix} -\frac{a}{b} - \frac{c}{b} & -\frac{a}{b} + \frac{c}{b} \\ \frac{1}{2} \left(1 - \frac{ac}{r^2}\right) & \frac{1}{2} \left(1 + \frac{ac}{r^2}\right) \end{pmatrix}$	$\begin{pmatrix} -\frac{a}{b} - \frac{r}{b} & -\frac{a}{b} + \frac{r}{b} \\ \frac{1}{2} \left(1 - \frac{a}{r}\right) & \frac{1}{2} \left(1 + \frac{a}{r}\right) \end{pmatrix}$
	$\beta^2 \geq c$	$\beta^2 < c$
(1.2)	$\begin{pmatrix} \mu - \frac{c}{\beta} & \mu + \frac{c}{\beta} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} \mu - \beta & \mu + \beta \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

We note that the same reasoning applies to most of the other commonly used optimality criteria (A-, Elfving's minimax, Kiefer's Φ_p -criteria), independent whether they are used in a locally, Bayesian or sequential setup. As a representative example of Kiefer's class of Φ_p -criteria consider the A-optimal design for the logistic regression model (2.1) with parameterization (1.2). It is relatively straightforward to show that the A-optimal design for this model has equal masses at the points $\mu \pm v/\beta$, where v is the unique positive root of the equation

$$v + \frac{v^3}{\beta^4} = 2 \frac{e^v + 1}{e^v - 1}. \quad (2.4)$$

If $\beta \rightarrow 0$ this design converges to the measure with equal masses at the points $\mu - \sqrt{2}$, $\mu + \sqrt{2}$. Some numerical results for A-optimal designs in the logistic regression are presented in table 2.2 which also include the efficiencies for estimating the parameters μ and β . We observe that for $\beta \leq 0.5$ the A-optimal design is inefficient for estimating the slope β in the logistic regression but has excellent efficiencies for estimating the LD50. Obviously such a design should not be used for a statistical inference in a logistic regression.

For the sake of completeness we finally provide the A-optimal designs for the logistic regression model with parameterization (1.1). Here the A-optimal designs are given by

$$\xi_a = \begin{pmatrix} -\frac{a}{b} - \frac{v^*}{b} & -\frac{a}{b} + \frac{V^*}{b} \\ 1 - w^* & w^* \end{pmatrix}, \quad (2.5)$$

where the weight w^* is defined as $1/2$ if $a = 0$ and otherwise as $w^* = w(v^*)$, where

$$w(v) = 1/2 + \frac{a^2 + b^2 + v^2 - \sqrt{(a^2 + b^2 + v^2)^2 - 4a^2v^2}}{4av}$$

and v^* minimizes the function

$$f_A(v) = \frac{(a+v)^2 + b^2 - 4avw(v)}{h(v)v^2w(v)(1-w(v))}$$

The A-optimal designs for some values of a and b are listed in table 2.3. Again we observe that for small values of the slope parameter the A-optimal design becomes inefficient for estimating b .

Finally it is worthwhile to mention the particular role of the D-optimality criterion in Kiefer's class ($p = 0$), for which both parameterizations yield the same optimization problem in (2.2) and (2.3). For this criterion the optimal design has equal masses at the points $-a/b \mp v/b$ (for parameterization (1.1)) or $\mu \mp v/\beta$ (for parameterization (1.2)), where $v > 0$ is the unique solution of $(1+v) + e^v(1-v) = 0$, that is $v \approx 1.5434$. The locally D-optimal designs degenerate to one-point designs at the LD50 only for large values of the slope parameter. However, this corresponds to intuition because for a large slope in a Binary response one should take all observations close to the LD50.

Table 2.2. A-optimal designs for the logistic regression model with parameterization (1.2). The A-optimal design has equal mass at the points $\mu \pm v/\beta$. The third and fourth column show the efficiencies for estimating the individual parameters μ and β .

β	v	eff_μ	eff_β
0.01	0.0141	0.9999	0.0001
0.1	0.1413	0.9950	0.0113
0.2	0.2819	0.9804	0.0443
0.5	0.6925	0.8891	0.2427
1	1.3002	0.6731	0.6477

Table 2.3. A-optimal designs for the logistic regression model with parameterization (1.1). The A-optimal design puts masses w^* and $1 - w^*$ at the points $-a/b + v^*/b$ and $-a/b - v^*/b$. The last two columns show the efficiencies for estimating the individual parameters a and b .

a	b	v^*	w^*	eff_a	eff_b
0	0.01	0.1413	0.5	0.9950	0.0113
0	0.1	0.4435	0.5	0.9524	0.1066
0	1	1.3002	0.5	0.6731	0.6477
0	10	2.3146	0.5	0.3273	0.9982
1	0.01	1.0064	0.9941	0.9958	0.0106
1	0.1	1.0618	0.9461	0.9596	0.0999
1	1	1.4823	0.7068	0.7088	0.6255
1	10	2.3160	0.5109	0.3559	0.9977
10	0.01	2.3994	0.6200	1.0000	0.9424
10	0.1	2.3993	0.6200	1.0000	0.9424
10	1	2.3983	0.6187	1.0000	0.9437
10	10	2.3755	0.5586	0.9854	0.9861

3. Standardized optimal designs for Binary response models

The phenomena described in section 2 are caused by the different size of the elements in the Fisher information matrix. In order to avoid such problems Dette (1997) proposed for linear models the application of standardized optimality criteria which take into account different scales of the explanatory variables. We will use a similar approach for the Binary response model and standardize the elements in the Fisher information matrix by the best "standard deviations" of the estimators for the individual parameters obtainable by the choice of a design. To be precise, consider the Binary response model with parameterization (1.2) and let ξ_1 and ξ_2 denote the optimal designs for estimating μ and β , respectively, that is

$$\rho_j^2 = \inf_{\xi} e_j^T I_2^-(\xi, \theta) e_j = e_j^T I_2^-(\xi_j, \theta) e_j, \quad j = 1, 2, \quad (3.1)$$

where $e_1 = (1, 0)^T$, $e_2 = (0, 1)^T$ and the infimum is taken over all measures such that $e_j^T \theta$ is estimable. Define $\Delta = \text{diag}(\rho_1, \rho_2)$; for a design ξ the matrix

$$\tilde{I}_2(\xi, \theta) = \Delta I_2(\xi, \theta) \Delta \quad (3.2)$$

is called *standardized information matrix* for the Binary response model with parameterization (1.2). A *standardized Φ -optimal design* maximizes an optimality criterion Φ of $\tilde{I}_2(\xi, \theta)$, where Φ is an information function in the sense of Pukelsheim (1993), p. 119, that is a nonnegative, positive homogeneous, concave and upper semi-continuous function on the set of nonnegative definite matrices. In order to distinguish standardized

and commonly used optimality criteria we call the last-mentioned functions "classical" optimality criteria and the corresponding designs "classical" Φ -optimal designs. For example, the classical and standardized D-optimal designs coincide which follows directly from (3.2) and reflects the particular role of the determinant criterion. The discussion surrounding equations (2.2) and (2.3) shows that the standardized Φ -optimal design can be obtained by maximizing the function Φ of the matrix

$$\tilde{I}_2(\eta, \theta) = \int h(z) \begin{pmatrix} \beta^2 \rho_1^2 & -z \rho_1 \rho_2 \\ -z \rho_1 \rho_2 & z^2 \rho_2^2 / \beta^2 \end{pmatrix} d\eta(z) \quad (3.3)$$

and transforming the design via $x = \mu + z/\beta$. Note that the elements of a generalized inverse of $\tilde{I}_2(\xi, \theta)$ are given by $(e_i^T I_2^{-1}(\xi, \theta) e_j) / \rho_i \rho_j$, $(i, j = 1, 2)$, which explains the term "standardized" information matrix.

We propose this standardization in order to make all elements of the Fisher information matrix comparable (in the sense that they are of the same scale with respect to different values of the parameter θ). On the one hand this gives a better interpretation of a given optimality criterion (because it can never be dominated by a particular element of the Fisher information matrix), on the other hand this standardization will lead to a *canonical form* of the design problem which is similar to an approach of Ford, Torsney and Wu (1992) used for the c-optimality criterion. To see this we need the optimal designs for estimating the individual coefficients in the Binary response model, more precisely the "variances" produced by these designs. Throughout the paper we will assume that

$$\max_{t \in \mathbb{R}} h(t) = h(0), \quad (3.4)$$

then it follows from Elfving's theorem that

$$\rho_1^2 = \frac{1}{\beta^2 h(0)}. \quad (3.5)$$

Similarly, we obtain

$$\rho_2^2 = \beta^2 k^2, \quad (3.6)$$

where

$$k^{-2} = \max_{z \in \mathbb{R}} z^2 h(z) = c^2 h(c). \quad (3.7)$$

We assume here that the maximum in (3.7) exists and note that for most of the commonly used Binary response models the value $c > 0$ where the maximum in (3.7) is attained, is unique and can be obtained from table 4 in Ford, Torsney and Wu (1992). Summarizing this discussion we state the following result.

THEOREM 3.1. Consider a Binary response model with parameterization (1.2) such that (3.4) is satisfied and such that the maximum in (3.7) exists. The standardized Φ -optimal design ξ for this model is given by $\xi(x) = \eta(\beta(x - \mu))$, where the design η maximizes the canonical form

$$\Phi \left[\int h(z) \begin{pmatrix} 1/h(0) & -zk/\sqrt{h(0)} \\ -zk/\sqrt{h(0)} & z^2 k^2 \end{pmatrix} d\eta(z) \right], \quad (3.8)$$

which is independent of the parameter (μ, β) .

For a similar analysis of standardized optimal designs for Binary response models with parameterization (1.1) we assume that the curve

$$S = \left\{ \sqrt{h(w)} \begin{pmatrix} 1 \\ w \end{pmatrix} \mid w \in \mathbb{R} \right\} \quad (3.9)$$

is convex and that

$$\lim_{|w| \rightarrow \infty} w^2 h(w) = 0, \quad (3.10)$$

which is a common assumption for the determination of optimal designs in Binary response models (see Wu, 1988). The optimal designs for estimating the individual parameters a and b can be constructed with the canonical form approach proposed by Ford, Torsney and Wu (1992). This reduces the problem to the determination of the c -optimal designs in the weighted linear regression $(\gamma + \delta w)\sqrt{h(w)}$ for $c = \tilde{e}_1 = (1, a)^T$ and $c = \tilde{e}_2 = (0, b)^T$. By an application of Elfving's theorem we thus obtain

$$\rho_2^2 = \inf_{\xi} e_2^T I_1^{-1}(\xi, \theta) e_2 = \frac{b^2}{c^2 h(c)} \quad (3.11)$$

$$\rho_1^2 = \inf_{\xi} e_1^T I_1^{-1}(\xi, \theta) e_1 = \frac{m^2}{h(c)} \quad (3.12)$$

where c is defined by (3.7) and

$$m^2 = m^2(a) = \begin{cases} \frac{a^2}{c^2} & \text{if } |a| > c \\ \frac{h(c)}{h(a)} & \text{if } |a| \leq c \end{cases} \quad (3.13)$$

The standardized information matrix for a binary response model with parameterization (1.1) is calculated from (2.2), (3.2), (3.11) and (3.12), that is

$$\bar{I}_1(\eta, \theta) = \Delta I_1(\eta, \theta) \Delta = \int \frac{h(w)}{h(c)} \begin{pmatrix} m^2 & \frac{w-a}{c} m \\ \frac{w-a}{c} m & \frac{(w-a)^2}{c^2} \end{pmatrix} d\eta(w) \quad (3.14)$$

Note that this matrix is independent of the scaling parameter b but depends on the location a . The standardized Φ -optimal design ξ is obtained from $\xi(x) = \eta(a + bx)$ where η maximizes the canonical form $\Phi(\bar{I}_1(\eta, \theta))$.

The discussion so far has shown how the standardized optimal design problem can be reduced to the optimization of a canonical form similar as proposed by Ford, Torsney and Wu (1992) for the (classical) D- and c-criterion. Note that the approach of these authors cannot be transferred to other classical optimality criteria, whereas the canonical form derived here applies to all standardized optimality criteria. In some cases classical optimal designs are known from the literature and the standardized optimal designs can easily be determined from these measures by the following corollary. The proof is an immediate consequence of (2.2), (3.14), (2.3) and theorem 3.1.

COROLLARY 3.2.

- (a) Let $\xi_{a,b}$ denote the classical Φ -optimal design for a Binary response model with parameterization (1.1) satisfying (3.9) and (3.10). The standardized

Φ -optimal design $\tilde{\xi}$ for this model is given by

$$\tilde{\xi}(x) = \xi_{a,s} \left\{ \frac{bx}{s} \right\}$$

where $s = mc$ and m is defined in (3.13).

- (b) Let $\xi_{\mu,\beta}$ denote the classical Φ -optimal design for a Binary response model with parameterization (1.2) such that (3.4) is satisfied and such that the maximum in (3.7) exists. The standardized Φ -optimal design for this model is given by

$$\tilde{\xi}(x) = \xi_{0,s} \left\{ \frac{\beta}{s}(x - \mu) \right\}$$

where $s = 1/\sqrt{k\sqrt{h(0)}}$ and k is defined by (3.7).

REMARK 3.3. Note that the standardized optimal designs are invariant with respect to scale changes of the explanatory variable in the following sense. If the explanatory variable x is scaled with a constant $c > 0$ (caused for example by changing the units of measurements) the parameter (μ, β) in model (1.2) changes to $(c\mu, \beta/c)$. By theorem 3.1 the standardized Φ -optimal designs $\tilde{\xi}$ and $\tilde{\xi}_c$ for the parameters (μ, β) and $(c\mu, \beta/c)$ are related by $\tilde{\xi}(cx) = \tilde{\xi}_c(x)$. A similar analysis applies to Binary regression models with parameterization (1.1). This means that a scale change for x will not affect the standardized Φ -optimal design, except that it is scaled in the same way as the explanatory variable. Moreover, if parameterization (1.2) is used, the same arguments apply to location transformations. In other words, the standardized optimal designs are invariant with respect to affine

transformations, if parameterization (1.2) is used. This provides a strong advantage over the alternative parameterization (1.1).

4. Classical and standardized optimal designs for the logit and the probit model

In this section we will illustrate the calculation of standardized optimal designs for the logistic regression (2.1) and the probit model. As pointed out in section 3, the standardized optimal designs can be calculated by using the canonical form in (3.8) or (3.14), or can be obtained from the classical optimal designs for specific parameter values.

4.1 Logistic regression

We first illustrate the calculation of the standardized A-optimal design for parameterization (1.2), which can be obtained from corollary 3.2 and the discussion at the end of section 2. We note that $s \approx 1.15130$ and find that the standardized A-optimal design has equal masses at the points $\mu \mp v_A/\beta$, where v_A is the unique positive solution of (2.4) with $\beta = s$, that is $v_A \approx 1.45655$. The efficiencies of this design for estimating the individual parameters do not depend on μ and β and are 0.6131 (for μ) and 0.7403 (for β), respectively.

For the calculation of the standardized E-optimal designs in the logistic regression we can use the results listed in table 2.1. Observing that the constant c in (3.7) is defined as the unique positive solution of

$$2 + x + 2e^x - xe^x = 0 \quad (4.1)$$

the standardized E-optimal design for the parameterization (1.2) is obtained from table 2.1 by an application of corollary 3.2 (note that by (3.4) and (3.7) $s^2 = 1/k\sqrt{h(0)} = c\sqrt{h(c)/h(0)} < c$) and puts equal masses at the points $\mu \mp s^2/\beta$, where $s^2 \approx 1.32549$ can be drawn from table 4 in Ford, Torsney and Wu (1992). The efficiencies of this design for estimating the individual parameters are equal and given by 0.6633.

The parameterization (1.1) is slightly more complicated because two cases have to be treated separately. For example, consider the case $|a| \leq c$, where c is the unique positive solution of (4.1). The equality $a^2 + c^2 m^2 = c^2$ is equivalent to

$$a^2 + \frac{c^2 h(c)}{h(a)} = c^2 \quad (4.2)$$

which has a unique positive solution a_0 is given by $a_0 \approx 1.61022$. Consequently, if $a_0 \leq |a| \leq c$ (which implies $a^2 + c^2 m^2 \geq c^2$) we obtain from table 2.1 and corollary 3.2 that the standardized E-optimal design for a logistic regression with parameterization (1.1) has masses

$$\frac{1}{2} \left(1 \pm \frac{c/a}{1 + \rho^2(c, a)} \right)$$

at the points $(-a \pm c)/b$ where

$$\rho(c, a) = \frac{c}{a} \sqrt{\frac{h(c)}{h(a)}}. \quad (4.3)$$

The standardized E-optimal designs for the remaining cases $|a| \leq a_0$ (which implies $a^2 + c^2 m^2 \leq c$) and $|a| > c$ (which implies $a^2 + c^2 m^2 = 2a^2 > c^2$) are obtained in the same way and listed in table 4.1.

Table 4.1. Standardized E-optimal designs for the logistic regression model (2.1) using the two different parameterizations (1.1) and (1.2). Here c and a_0 are the positive solutions of (4.1) and (4.2), respectively ($c \approx 2.39936$, $a_0 \approx 1.61022$) and $\rho(c, a)$ is defined in (4.3).

(1.1)	$ a < a_0$	$\begin{pmatrix} -\frac{a}{b} - \frac{a\sqrt{1+\rho^2(c, a)}}{b} & -\frac{a}{b} + \frac{a\sqrt{1+\rho^2(c, a)}}{b} \\ \frac{1}{2} \left(1 - \frac{1}{\sqrt{1+\rho^2(c, a)}} \right) & \frac{1}{2} \left(1 + \frac{1}{\sqrt{1+\rho^2(c, a)}} \right) \end{pmatrix}$
	$a_0 \leq a \leq c$	$\begin{pmatrix} -\frac{a}{b} - \frac{c}{b} & -\frac{a}{b} + \frac{c}{b} \\ \frac{1}{2} \left(1 - \frac{c/a}{1+\rho^2(c, a)} \right) & \frac{1}{2} \left(1 + \frac{c/a}{1+\rho^2(c, a)} \right) \end{pmatrix}$
	$c < a $	$\begin{pmatrix} -\frac{a}{b} - \frac{c}{b} & -\frac{a}{b} + \frac{c}{b} \\ \frac{1}{2} \left(1 - \frac{c}{2a} \right) & \frac{1}{2} \left(1 + \frac{c}{2a} \right) \end{pmatrix}$
(1.2)		$\begin{pmatrix} \mu - \frac{c}{b} \sqrt{\frac{h(c)}{h(0)}} & \mu + \frac{c}{b} \sqrt{\frac{h(c)}{h(0)}} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

Note that the standardized E-optimal designs never become inefficient for estimating the individual parameters. The efficiencies are functions of the intercept $|a|$; for the first case ($|a| < a_0$), the efficiencies for a and for b are equal and decrease from $\text{eff}_a = \text{eff}_b = 0.6634$ for $a = 0$ to $\text{eff}_a = \text{eff}_b = 0.5496$ for $|a| = a_0$. In the second case ($a_0 \leq |a| \leq c$), the efficiencies are nearly equal and increase to $\text{eff}_a = \text{eff}_b = 3/4$ for $|a| = c$. In the third case ($|a| > c$), the efficiencies are equal again, increasing and explicitly given by $1 - \frac{c^2}{4a^2}$.

The standardized A-optimal designs for the logit model with parameterization (1.1) are obtained in the same manner from the classical ones (2.5) and corollary 3.2. The details are omitted for the sake of brevity. Again the standardized A-optimal design is never inefficient for the inference on the individual parameters. More precisely, both eff_a and eff_b are increasing functions of $|a|$ and bounded below by $\text{eff}_a = 0.6131$ and $\text{eff}_b = 0.7403$ where this bound is attained at $|a| = 0$.

4.2 Probit model

To demonstrate that the described phenomena do not depend on the special choice of the logistic regression model, but are actually shared by all Binary response models considered in sections 2 and 3, we now investigate the probit model. For the sake of brevity we will concentrate on the E-optimality criterion and on parameterization (1.2). Note that this parameterization has the important advantage over the alternative parameterization (1.1) that the resulting optimal designs are invariant with respect to affine transformations

(see the discussion at the end of section 3). As pointed out in section 3 the standardized E-optimal designs can easily be obtained from their classical counterparts by choosing special values for the parameter θ . We will therefore start with the calculation of the classical E-optimal designs for the probit model. Here the function h is given by

$$h(z) = \frac{\varphi^2(z)}{\phi(z)(1 - \phi(z))} \quad z \in \mathbb{R},$$

where ϕ is the standard normal distribution function and φ its derivative, the normal density. The quantities c and k in (3.7) are obtained numerically as

$$c \approx 1.57504, \quad k = \left(c\sqrt{h(c)}\right)^{-1} \approx 1.28203$$

It is well known (see Pukelsheim, 1993, p. 190) that in regression models with two parameters there always exists a locally optimal design with at most three support points. Due to the symmetry of parameterization we can restrict ourselves to symmetric designs with respect to the LD50 μ . In the appendix we show that for the probit model in this parameterization any symmetric optimal design with respect to any information function in the sense of Pukelsheim (1993) has at most two support points.

4.2(a) Classical E-optimal designs

Standard results show that there exists a symmetric optimal design. By the proposition from the appendix and the fact that any E-optimal design has at least as many support points as the number of parameters in the model, it suffices to consider designs of the form

$$\lambda_v = \begin{pmatrix} \mu - v/\beta & \mu + v/\beta \\ 1/2 & 1/2 \end{pmatrix}, \quad v > 0. \quad (4.6)$$

For the corresponding minimum eigenvalue we obtain

$$\gamma_{\min}(I_2(\eta_v, \theta)) = \begin{cases} \beta^2 h(v) & \text{if } \beta^2 \leq v^2 \\ v^2/\beta^2 h(v) & \text{if } \beta^2 > v^2 \end{cases} \quad (4.7)$$

Since h is a decreasing function on the positive real numbers and $v^2 h(v)$ is increasing for $v < c$ and decreasing for $v > c$, this expression is maximized for

$$v = \begin{cases} \beta & \text{if } \beta^2 \leq c \\ c & \text{if } \beta^2 > c \end{cases} \quad (4.8)$$

and the resulting E-optimal designs are listed in table 4.2. It is worth mentioning that the E-optimal designs for the probit model show exactly the same structure as the E-optimal designs for the logistic regression. In fact for $\beta^2 > c$ the classical E-optimal designs for every Binary response model (1.2) satisfying the assumptions of theorem 3.1 have this structure, and for $\beta^2 < c$ the E-optimal symmetric two-point design is of this form (which follows readily by repeating the above arguments). Note, however, that the constant c in both tables is defined differently because the two models yield different functions h in the defining equation (3.7).

Table 4.2. Classical E-optimal designs for the probit model with parameterization (1.2). The constant c is defined as the positive solution of (3.7) [$c \approx 1.57504$].

$\beta^2 > c$	$\beta^2 \leq c$
$\begin{pmatrix} \mu - \frac{c}{\beta} & \mu + \frac{c}{\beta} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} \mu - \beta & \mu + \beta \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

Note that in the probit model we observe the same difficulties as in the logistic regression model. The smaller the value of the slope parameter β , the

closer the support points of the E-optimal design are to the LD50, instead of (as intuition would advise) covering a larger range of dose levels. Therefore the design becomes inefficient for inference on the slope parameter β . If, for example, $\beta = 0.1$, the E-optimal design has efficiencies of 99.99% (for μ) and 0.01% (for β).

4.2(b) Standardized E-optimal designs

As pointed out in section 3 the standardized E-optimal design for the probit model is obtained from the classical (locally) E-optimal designs and the second part of corollary 3.2. It turns out that it puts equal masses at the points $\mu \pm v_E/\beta$ where $v_E = s^2 \approx 0.9776$, analogously to the logit model.

Note that the standardized E-optimal design is never inefficient for estimating the individual parameters irrespective of the value of the "true" parameters μ and β . This can be seen by calculating the inverse Fisher information matrix for the design given above and forming the ratio with the best values in (3.6) and (3.5). For the probit model with parameterization (1.2) the standardized E-optimal design has equal efficiencies of 70.1% for μ and β .

5. Conclusion

In this paper we illustrated some practical aspects of locally optimal designs in Binary response models with two parameters with respect to the commonly used optimality criteria. As representative examples we considered the A- and E-criterion for the logistic and the probit regression model. It is demonstrated that these designs should not be used for small values of the slope, because in this case they degenerate to one-point designs which are inefficient for the

statistical inference of all parameters in the model. Since in every Binary response model the slope parameter can be diminished by changing the unit of measurement appropriately, we do not recommend these criteria for the construction of optimal designs in such models.

In order to avoid these difficulties the concept of a standardized Fisher information matrix is used, which eliminates different scales of the elements of the Fisher information matrix, in Binary response models mainly caused by different sizes of the parameters. The standardized locally optimal designs can be obtained by either maximizing a canonical form of the optimality criterion or from the classical locally optimal designs by choosing specific values for the parameters. The resulting standardized optimal designs allow an efficient estimation of all parameters in the model.

An alternative way to produce a canonical form of the design problem was proposed by Sitter and Wu (1993) who used the parameterization $(x - \mu)/\beta$ in the binary response model with two parameters. The optimal designs with respect to this parameterization can easily be obtained from the classical optimal designs for the parameterization (1.2). For example, in both the logistic and the probit regression model the E-optimal design has equal masses at the points $\mu \mp 1/\beta$. We note, however, that this approach uses a different interpretation of the slope parameter.

Although we restricted our investigation mainly to locally optimal designs and the E-criterion, all results can be transferred to other optimality criteria depending on the Fisher information matrix (e.g. Kiefer's Φ_p -criteria, Elfving's minimax criterion) and other approaches to the optimal design problem for a Binary response model (e.g. Bayesian or sequentially optimal designs). In

all cases a standardization will eliminate the different scales in the Fisher information matrix (caused by the different size of parameters) and will yield a canonical form of the optimization problem which does not depend on the slope parameter.

6. References

- ABDELBASIT, K.M. & PLACKETT, R.L. (1983). Experimental designs for binary data. *Journ. Amer. Statist. Assoc.*, **78**, 90–98.
- CHALONER, K. & LARNTZ, K. (1989). Optimal Bayesian experimental design applied to logistic regression. *J. Statist. Plann. Inf.*, **21**, 191–208.
- CHERNOFF, H. (1953). Locally optimal designs for estimating parameters. *Ann. Math. Statist.*, **24**, 586–602.
- DETTE, H. (1997). Designing of experiments with respect to “standardized” optimality criteria. *J.R. Statist. Soc., Ser. B.*, **59**, 97–110.
- DETTE, H. & HAINES, L.M. (1994). E-optimal designs for linear and nonlinear models with two parameters. *Biometrika*, **81**, 739–754.
- ELFVING, G. (1952). Optimum allocation in linear regression theory. *Ann. Math. Statist.*, **23**, 255–262.
- FORD, I. & SILVEY, S.D. (1980). A sequentially constructed design for estimating a nonlinear parametric function. *Biometrika*, **67**, 381–388.
- FORD, I., TORSNEY, B. & WU, C.F.J. (1992). The use of a canonical form in the construction of locally optimal designs for nonlinear problems. *J.R. Statist. Soc. B.*, **54**, 569–583.
- HAINES, L.M. (1992). Optimal design for inverse quadratic polynomials. *South African Statist. J.*, **26**, 25–41.
- HAINES, L.M. (1993). Optimal design for nonlinear regression models. *Commun. Statist. A*, **22**, 1613–1672.

- HAINES, L.M. (1995). A geometric approach to optimal design for one-parameter nonlinear models. *J.R. Statist. Soc., B*, **57**, 575–598.
- JOHNSON, N.I. & KOTZ, S. (1970). *Continuous univariate distributions –2*. Houghton Mifflin, Boston.
- KALISH, L.A. (1990). Efficient design for estimation of median lethal dose and quantal dose-response curves. *Biometrics*, **46**, 37–48.
- KIEFER, J. (1974). General equivalence theory for optimal designs (approximate theory). *The Annals of Statistics*, **2**, 849–879.
- MINKIN, S. (1987). Optimal design for binary data. *J. Amer. Statist. Assoc.*, **82**, 1098–1103.
- MUKHOPADHYAY, S. & HAINES, L.M. (1995). Bayesian D-optimal designs for the exponential growth model. *J. Statist. Planng. Infer.*, **44**, 385–397.
- PUKELSHEIM, F. (1993). *Optimal design of experiments*. Wiley, New York.
- SITTER, R.R. & WU, C.F.J. (1993). Optimal designs for binary response experiments: Fieller-, D- and A-criteria. *Scand. J. Statist.*, **20**, 329–341.
- WU, C.F.J. (1985). Asymptotic inference from sequential design in a nonlinear situation. *Biometrika*, **72**, 553–558.
- WU, C.F.J. (1988). Optimal design for percentile estimation of a quantal response curve. In *optimal design and analysis of experiments*, Ed. Y. Dodge, V.V. Fedorov & H.P. Wynn, pp. 213–223. North Holland, Amsterdam.

7. Appendix

PROPOSITION. Let Ψ be an information function in the sense of Pukelsheim (1993). Then any symmetric Ψ -optimal design η_Ψ for the probit model with parameterization (1.2) has at most two support points.

PROOF. Observing the discussion in section 2 we have to maximize $\Psi(I_2(\eta, \theta))$, where $I_2(\eta, \theta)$ is defined in (2.3). The general equivalence theorem in Pukelsheim (1993), p. 177, shows that η_Ψ maximizes $\Psi(I_2(\eta, \theta))$ if and only if

$$h(w) \left(\beta, \frac{w}{\beta} \right) N(\eta_\Psi) \left(\beta, \frac{w}{\beta} \right)^T \leq 1 \quad (\text{A1})$$

for all $w \in \mathbb{R}$, where $N(\eta_\Psi)$ is a suitably chosen, nonnegative definite matrix, and there is equality at the support points of every Ψ -optimal design. Moreover, since the matrix $I_2(\eta, \theta)$ is diagonal the matrix $N(\eta_\Psi)$ can be chosen diagonal, too. The inequality (A1) can easily be rewritten as

$$g(w) := h(w)[\alpha + \gamma w^2] \leq 1, \quad (\text{A2})$$

where $\alpha, \gamma \geq 0$ are defined appropriately. We prove that the derivative of g has at most three zeros which implies that the design η_Ψ has at most two support points (where g attains local maxima). The derivative g' is given by

$$g'(w) = h(w) \left[2\gamma w + (\alpha + \gamma w^2) \left(\frac{2\phi(w) - 1}{\phi(w)} \frac{\varphi(w)}{1 - \phi(w)} - 2w \right) \right].$$

We show (a) that $g'(w)/h(w) < 0$ for all $w \geq \sqrt{3}$ and (b) that g'/h is a concave function on $(0, \sqrt{3})$. This implies that g' has at most one positive zero (which must be located in the interval $(0, \sqrt{3})$) and the symmetry of g then completes the proof.

(a) For $w > 0$ we have the inequality

$$\left(2 - \frac{1}{\phi(w)} \right) \frac{\varphi(w)}{1 - \phi(w)} < \frac{\varphi(w)}{1 - \phi(w)} < w + \frac{1}{w}$$

(see for example Johnson/Kotz (1970), chapter 33). Inserting this into (A3) shows $g'(w)/h(w) < 0$ for $w > \sqrt{3}$.

(b) The key to the proof is the concavity of the function

$$v(w) = \left(2 - \frac{1}{\phi(w)}\right) \frac{\varphi(w)}{1 - \phi(w)} - 2w$$

for $0 < w \leq (0, \sqrt{3}]$, which can easily be seen in figure 1 below where the second derivative of v is plotted. A rigorous proof of this assertion, which requires more precise bounds of Mill's ratio $\frac{1 - \phi(w)}{\varphi(w)}$ and Taylor expansions of $\phi(w)$ up to order 10, is omitted for the sake of brevity. The concavity of g'/h is now obtained by straightforward computation of its second derivative observing $v'(0) = 4/\pi - 2 < 0$, $v(0) = 0$ and $\alpha, \gamma \geq 0$.

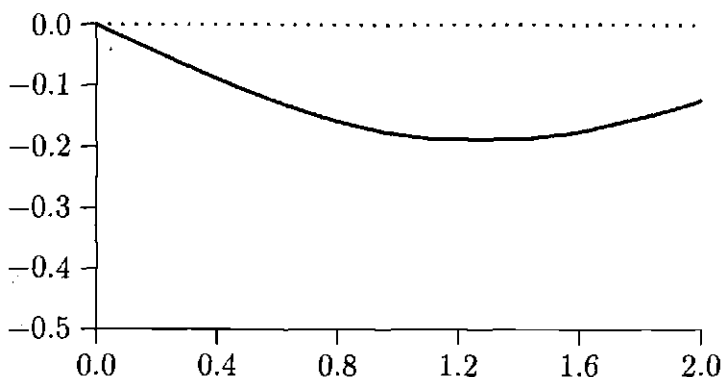


Figure 1. The second derivative of the function $v(w) = (2 - 1/\phi(w))\varphi(w)/(1 - \phi(w)) - 2w$ in the interval $[0, 2]$.